
Unit 1

Lesson 10

Equations and inequalities

Of the first degree in one variable in R

1. Equation

$$\text{Ex) } \frac{1}{2}x - \sqrt{5} = 0 \quad \frac{1}{2}x = \sqrt{5} \quad x = 2\sqrt{5}$$

* Equations on the first degree

* x is a variable (unknown)

Find in R the S.S:

1. $7y - \sqrt{7} = 6\sqrt{7}$

2. $x - \sqrt{5} = 1$

Solution

1. $7y - \sqrt{7} = 6\sqrt{7}$

$$7y = 6\sqrt{7} + \sqrt{7}$$

$$7y = 7\sqrt{7}$$

$$y = \sqrt{7} \quad \text{s.s } \{\sqrt{7}\}$$

$$2. x - \sqrt{5} = 1$$

$$x = \sqrt{5} + 1 \quad \text{s.s } \{\sqrt{5} + 1\}$$

2- inequalities

$$\text{Ex) } 3x + 2 \leq 1$$

$$3x \leq -1$$

$$x \leq -\frac{1}{3}$$

* inequalities of the first degree

* x is unknown

Remark:

When multiply or divide the two sides of an inequality by a negative number we should Change the symbol of inequality.

* Find in R the S.S:

$$1. x - 2 \geq 3x - 5$$

$$2. x - 1 < 3x - 3 \leq X + 5$$

solution

$$1. x - 2 \geq 3x - 5 \quad 3 \geq 2x \quad \therefore \frac{3}{2} \geq x$$

$$x \text{ s.s }] -\infty, \frac{3}{2}]$$

$$2. x - 1 < 3x - 3 \leq x + 5$$

Add -x

$$-1 < 2x - 3 \leq 5$$

$$2, 2x \leq 8 \quad 1 < x \leq 4 \quad \text{s.s }]1, 4]$$

Question find the s.s and represent solution on the number line.

$$1- 2-\sqrt{6} x = |-8|$$

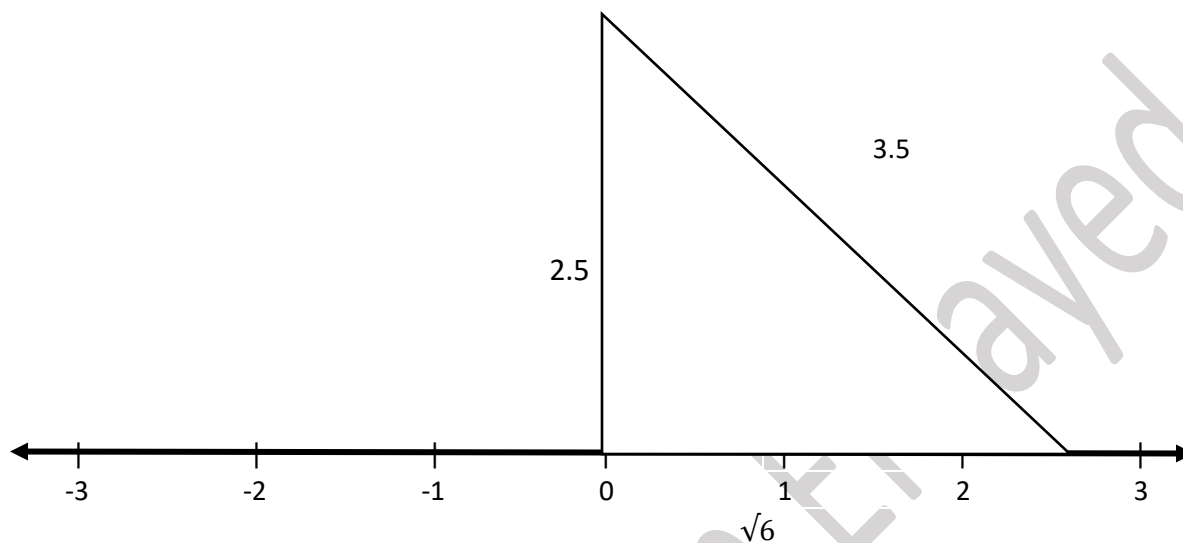
$$2- x + 2\sqrt{3} = 3$$

Solution

$$1. 2-\sqrt{6} x = |-8| \quad -\sqrt{6} x = 8 - 2$$

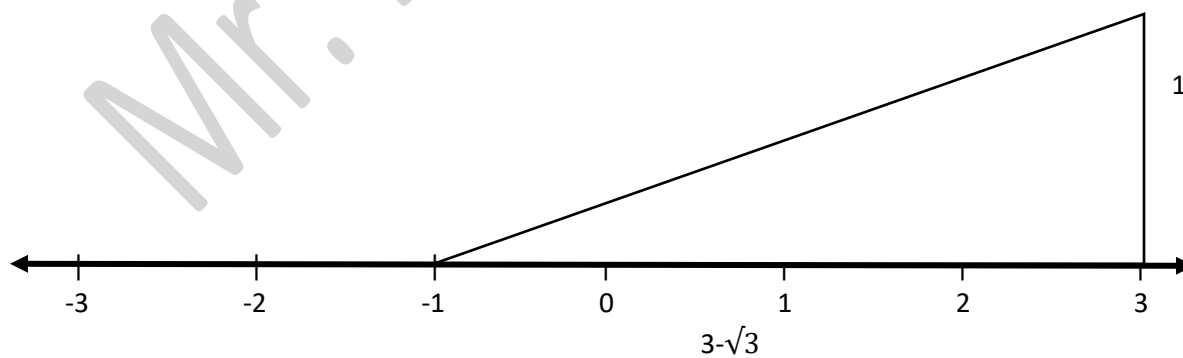
$$x = -\frac{6}{\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = = \frac{-6\sqrt{6}}{6} - \sqrt{6}$$

$$\text{s.s}\{-\sqrt{6}\}$$



$$2 - x + 2\sqrt{3} = 3$$

$$x = 3 - 2\sqrt{3} \quad \text{s.s } \{ 3 - 2\sqrt{3} \}$$



Question: find the s.s and represent on the number line

$$1- 1 < 5-x \leq 3$$

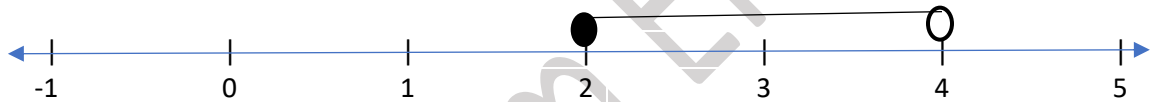
$$2- \sqrt[3]{-8} \leq x + 1 \leq \sqrt{9}$$

$$1. 1 < 5 - x \leq 3$$

$$- 4 < -x + 1 \leq -2$$

$$4 > x \geq 2$$

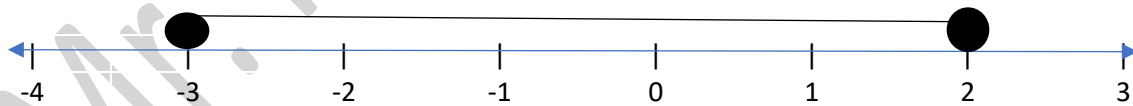
$$S.S[2, 4[$$



$$2. \sqrt[3]{-8} \leq x + 1 \leq \sqrt{9} \quad -2 \leq x + 1 \leq 3$$

$$-3 \leq x \leq 2$$

$$s.s[-3, 2]$$



H.W

Question 1: find the solution set, then represent the solution on the number line.

1. $x + 5 = 0$

2. $5x + 6 = 1$

3. $2x + 4 = 3$

4. $2x - 3 = 4$

5. $x + 2\sqrt{3} = 3$

6. $-x > 3$

7. $5 - x > 3$

8. $2x + 5 \geq 3$

9. $1 - 5x < 6$

10. $\frac{1}{2}x + 1 \leq 2$

11. $3 - 2x \leq 7$

Question 2: find the S.S

1. $|-3| < 2x - 1 < 5$

2. $5x - 3 < 2x + 9$

3. $4x \leq 5x + 2 < 4x + 3$

4. $x - 1 < 3x - 1 \leq x + 1$

5. $\sqrt{2}x \leq 4$